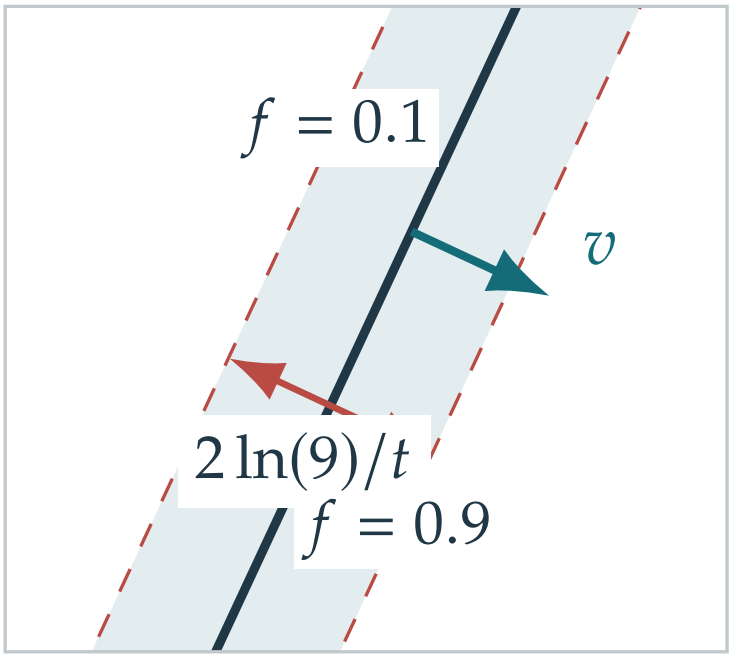
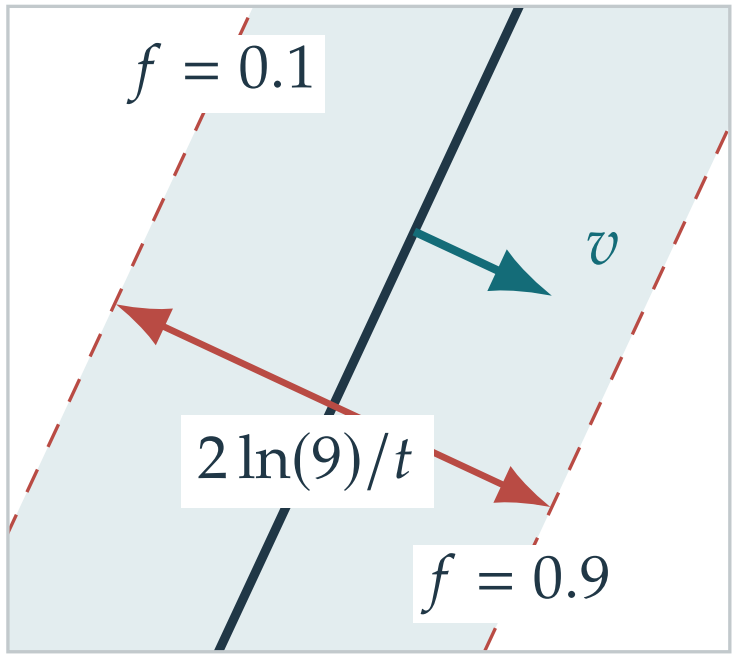
$\|\beta\| = 3$



$\|\beta\| = 1.4$



$$\beta = tv, \quad \|v\| = 1, \quad f(x, \beta) = \theta(t\langle v, x\rangle)$$

Same boundary $\langle v, x\rangle = 0$; width $2\ln(9)/t$.